

Yen-Hsiang (Robert) Huang

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Education

Ph.D., Electrical and Computer Engineering

Georgia Institute of Technology, Atlanta, GA GPA: 4.0/4.0

Sept. 2022 – Dec. 2027 (expected)

B.S., Electrical Engineering

National Taiwan University (NTU), Taipei, Taiwan GPA: 4.06/4.30

Sept. 2017 – July 2022

Experience

R&D Engineer Intern

Synopsys Inc.

May 2025 – Dec. 2025

Sunnyvale, CA

- Investigated machine-learning-based signal-integrity (SI) analysis for 2.5D ICs, evaluating multiple models to predict final-stage SI performance from early-stage design data.
- Independently implemented two provided prototypes in C++ and merged the code into the main branch.

Graduate Research Assistant

SCCAD (former GTCAD), Georgia Tech; Advisor: Prof. Sung Kyu Lim

Sept. 2022 – Present

3D IC Optimization

Fall 2024 – Present

- Timing optimization through reduced critical-path snaking and inter-die cell relocation (Snake-3D).
- Developed an RC-calibrated differentiable clock-tree optimization framework for 3D ICs that incorporates interconnect RC effects into timing-aware optimization (CLK-3D).

Die Bonding Bumps and Pads Legalization for 3D ICs

Sept. 2022 – May 2023

- Developed 3D-via legalization with refinement and minimal runtime overhead; eliminated all 3D-via overlaps when via utilization is below 40%.

SRC JUMP 2.0 CHIMES

Fall 2023 – Spring 2024

- Bridged the PPA gap between legacy-node 3D ICs and advanced-node 2D ICs: reduced WNS by 50–93% versus SOTA, reaching 87.9–97.7% of advanced-node 2D performance.

Software Engineer R&D Intern

Synopsys Inc. – Concurrent Clock-and-Data (CCD) Team

July 2021 – Aug. 2021

Hsinchu, Taiwan

- Enhanced the flip-flop merging algorithm in IC Compiler II (ICC II) by considering flip-flop timing (slack), improving worst/total negative slack by 5.14%/3.47% on 13 real industrial benchmarks.

Software Engineer R&D Intern

Cadence Inc. – Conformal Team

June 2020 – Sept. 2020

Hsinchu, Taiwan

- Provided engineering change order (ECO) test cases in a five-person team for Conformal's robustness testing.
- Generated 2,000+ cases across five categories in which Conformal acted unusually and proposed three improved flows to address them.

Independent Study, NTU

EDA Lab (Prof. Yao-Wen Chang); ALCom Lab (Prof. Jie-Hong Jiang)

Apr. 2020 – Sept. 2021

- Placed in the top five twice in three physical-design global contests (ISPD 2021, ICCAD 2021).
- Implemented a reversible circuit synthesizer for controlled-NOT (CNOT) gates.

Publications

- Y.-H. Huang** and S. K. Lim, "CLK-3D: RC-Calibrated Differentiable Clock-Tree Optimization for 3D ICs," IEEE/ACM International Conference on Computer-Aided Design, 2026.
- Y.-H. Huang** and S. K. Lim, "Snake-3D: Differentiable Learning for Cross-Tier Logic Path Snaking Optimization in 3D ICs," IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2025.
- Y.-H. Huang**, S. Pentapati, A. Agnesina, M. Brunion, and S. K. Lim, "On Legalization of Die Bonding Bumps and Pads for 3D ICs," IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2024.
- S. Pentapati, A. Agnesina, M. Brunion, **Y.-H. Huang**, and S. K. Lim, "On Legalization of Die Bonding Bumps and Pads for 3D ICs," ACM International Symposium on Physical Design (ISPD), 2023.

Skills and Honors

Programming: C/C++, Python, MATLAB, L^AT_EX

Honors: Honorable Mention (4th place worldwide), ACM ISPD Contest 2021 | TSMC Undergraduate Scholarship (about 20 recipients in Taiwan per semester), 2020–2021 | Dean's List, ranked 1st (4.30/4.30), Spring 2020